Day 4 Task — CyberChef Lab

Time needed	Deadline	Where to submit
45 minutes	Before the Day 5 session	Portal → Day 4 → Upload

Open **https://gchq.github.io/CyberChef** in your browser (no install, no account). Paste text into the Input box, drag operations from the left into the Recipe, and read the Output. This is the same tool analysts use to unpack real phishing payloads.

Part A — Decode the hidden messages (encoding ≠ encryption)

Each string below hides a readable sentence. Find the right operation(s), decode it, and write the sentence plus the operation you used.

A1. Hint: ends with '=' signs are a giveaway.

```
V2VsY29tZSB0byBUZWtsb29teS4gRW5jb2RpbmcgaXMgbm90IGVuY3J5cHRpb24u
```

A2. Hint: only the characters 0–9 and a–f.

```
486173682065766572797468696e6720796f7520646f776e6c6f61642e
```

A3. Hint: a Caesar shift with a famous number.

```
Mreb gb Qrsraqre
```

A4. Hint: two layers — undo A1's trick, then A2's.

```
NDI2MTYzNmI3NTcwNzMyMDYyNjU2MTc0MjA3MjYxNmU3MzZmNmQ3NzYxNzI2NQ==
```

#	Decoded message	Operation(s) used
A1		
A2		
A3		
A4		

Part B — Hashing: tiny change, totally different fingerprint

Use the SHA-256 operation. Hash each input and copy the first 8 characters of the output.

Input (type exactly)	First 8 chars of SHA-256
tekloomy	
Tekloomy	
tekloomy (with a space at the end)	

B4. One capital letter changed. How much of the hash changed? What does this tell you about why hashes are used to verify downloads? (2 lines)

Part C — Inspect a real certificate

In Chrome or Edge, go to **https://www.google.com**, click the icon left of the address bar → 'Connection is secure' → 'Certificate is valid' (Firefox: padlock → Connection secure → More information → View Certificate).

Field	What you found
Issued to (Common Name / domain)	
Issued by (Certificate Authority)	
Valid until (expiry date)	
Public key algorithm (e.g. RSA, ECC) and size	

C5. Now visit **https://expired.badssl.com**. What warning do you see, and what does it actually mean? Should you click 'Proceed'? (2–3 lines)

Part D — Encrypt for real

In CyberChef, use **AES Encrypt**: mode CBC, key *00112233445566778899aabbccddeeff* (hex), IV *000102030405060708090a0b0c0d0e0f* (hex). Input: your first name. Copy the output. Then use **AES Decrypt** with the same key and IV to get your name back.

D1. Paste the encrypted output (hex): _____

D2. Change one character of the key and try to decrypt. What happens? Why does this matter for ransomware? (2 lines)

How to submit

- One file: Google Doc, PDF, or a clear photo of this sheet filled in
- Must contain: Part A table, Part B table + B4, Part C table + C5, Part D answers
- Upload under Day 4 on the portal before the Day 5 session

Marks (out of 10)

Part	Marks
A — all four messages decoded with operations named	4
B — three hashes correct + B4 reasoning	2
C — certificate fields + C5 reasoning	2
D — encryption/decryption shown + D2 reasoning	2

Counts toward your lab journal (30% of the program grade). Solutions revealed at the start of Day 5.